Architecture Design Report

This document presents the architectural design of the system, including its context, logical structure, runtime b

1. System Requirements

System Name: ResuFit: Precision CV Matching System

Architecture Style: Microservices

Functional Requirements

- System Functions
- Detailed Functional Specification
- Objectives
- User Characteristics

Non-Functional Requirements

- Standard Compliance

  Description: The system MUST comply with relevant standards and regulations, ensuring accessibility and u

- Accessibility Constraints

  Description: The system SHALL provide an accessible and user-friendly interface, ensuring that individuals

- Scalability

  Description: The system MUST be scalable to accommodate a large number of users, ensuring that the sys

- Data Privacy Compliance

  Description: The system MUST comply with relevant data privacy regulations, ensuring the secure storage a

2. Formal Architecture Specification (ACME ADL)

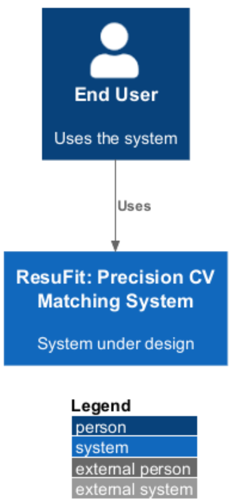
The following section provides a formal architectural specification using ACME ADL, capturing components, co

```
System ResuFit:PrecisionCVMatchingSystem {
  Property view : String = "runtime";
  Property architectureStyle : String = "Microservices";
  Property source : String = "AI";
  Component AssessmentModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Conducts precise assessments to identify challenges related
```

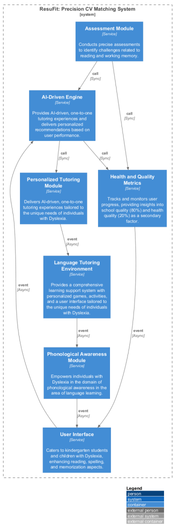
```
  Property state : String = "stateless";
  Property scaling : String = "horizontal";
}
Component PersonalizedTutoringModule = new Component {
  Port in;
  Port out;
  Property responsibility : String = "Delivers AI-driven, one-to-one tutoring experiences tailored
  Property state : String = "stateless";
  Property scaling : String = "horizontal";
}
Component LanguageTutoringEnvironment = new Component {
  Port in;
  Port out;
  Property responsibility : String = "Provides a comprehensive learning support system with person
  Property state : String = "stateless";
  Property scaling : String = "horizontal";
}
Component PhonologicalAwarenessModule = new Component {
  Port in;
  Port out;
  Property responsibility : String = "Empowers individuals with Dyslexia in the domain of phonolog
  Property state : String = "stateless";
  Property scaling : String = "horizontal";
}
Component UserInterface = new Component {
  Port in;
  Port out;
  Property responsibility : String = "Caters to kindergarten students and children with Dyslexia,
  Property state : String = "stateless";
  Property scaling : String = "horizontal";
}
Component AI-DrivenEngine = new Component {
  Port in;
  Port out;
  Property responsibility : String = "Provides AI-driven, one-to-one tutoring experiences and deli
  Property state : String = "stateless";
  Property scaling : String = "horizontal";
}
Component HealthandQualityMetrics = new Component {
  Port in;
  Port out;
  Property responsibility : String = "Tracks and monitors user progress, providing insights into s
  Property state : String = "stateless";
  Property scaling : String = "horizontal";
}
Attachment AssessmentModule.out to AI-DrivenEngine.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
  Property backpressure : String = "bounded";
}
Attachment AssessmentModule.out to HealthandQualityMetrics.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
  Property backpressure : String = "bounded";
}
Attachment AI-DrivenEngine.out to PersonalizedTutoringModule.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
```

```
    Property backpressure : String = "bounded";
  }
  Attachment PersonalizedTutoringModule.out to LanguageTutoringEnvironment.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
  }
  Attachment LanguageTutoringEnvironment.out to PhonologicalAwarenessModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
  }
  Attachment PhonologicalAwarenessModule.out to UserInterface.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
  }
  Attachment UserInterface.out to AI-DrivenEngine.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
  }
  Attachment AI-DrivenEngine.out to HealthandQualityMetrics.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
  }
  Attachment HealthandQualityMetrics.out to UserInterface.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
  }
  Property resilience_pattern : String = "Retry, CircuitBreaker";
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = false;
  Property authentication : String = "service-to-service";
  Property authorization : String = "RBAC";
}
```

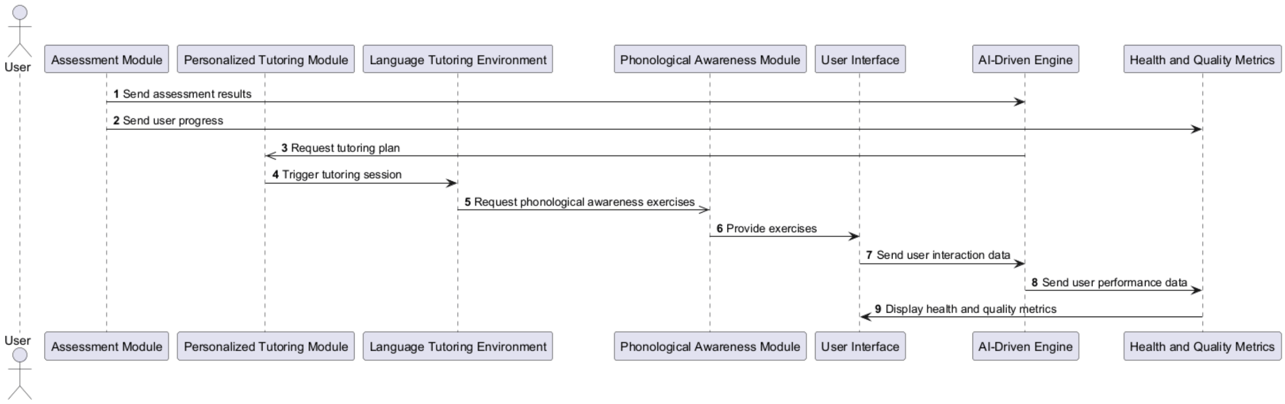
3. Context View



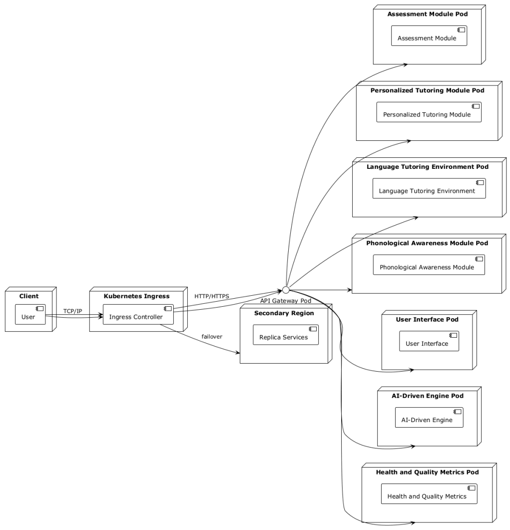
4. Logical View (C4 Container Diagram)



5. Process View (Runtime Interaction)



6. Physical View (Deployment Diagram)


Assessment Module Pod
Assessment Module
Personalized Tutoring Module Pod
Personalized Tutoring Module
Language Tutoring Environment Pod
Language Tutoring Environment
Phonological Awareness Module Pod
Phonological Awareness Module
Client
User
TCP/IP
Kubernetes Ingress
Ingress Controller
HTTP/HTTPS
API Gateway Pod
failover
Secondary Region
Replica Services
User Interface Pod
User Interface
AI-Driven Engine Pod
AI-Driven Engine
Health and Quality Metrics Pod
Health and Quality Metrics